wheelset-s2pinn: Phase 0 report (gate G0)

Generated from **results/** by experiments/make_g0_report.py. Gate file commit: 3b0f25f2d7858feaeaa7833f01447a45a558d964-dirty; generated 2026-10-11T14:28:13Z. Every number below is read from a results JSON written by an experiment script.

1. Gate summary

Item	Status	Key numbers
(a) S²-PINN reproduces stochastic diffusion within 2× of the paper's mean error	PENDING	needs a GPU run (experiments/colab_p0_reproduce.ipynb); CPU estimate 22.7 h
(b) SDOF shows a clear trend of required gPC degree with Π	PASS (trend clear)	Legendre: $p \propto \Pi^{0.91\pm0.01}$ (R^2=0.994); Hermite: overall $\Pi^{1.59}$, local slope → 1.93 at large Π
(c) Eq. (41) within 10% of measured effective damping, or measured relation reported	PENDING	
(d) Literature stop rule (method already published?)	NOT TRIGGERED	see docs/lit_review.md

2. (a) S²-PINN stochastic-diffusion reproduction

Pending. needs a GPU run (experiments/colab_p0_reproduce.ipynb); CPU estimate 22.7 h

CPU cost on this container (4 cores, no GPU): 4.09 s per step, i.e. 22.7 h for 20k steps. The 1 GB / 90 min cost limit refers to a T4 and cannot be checked on this hardware.

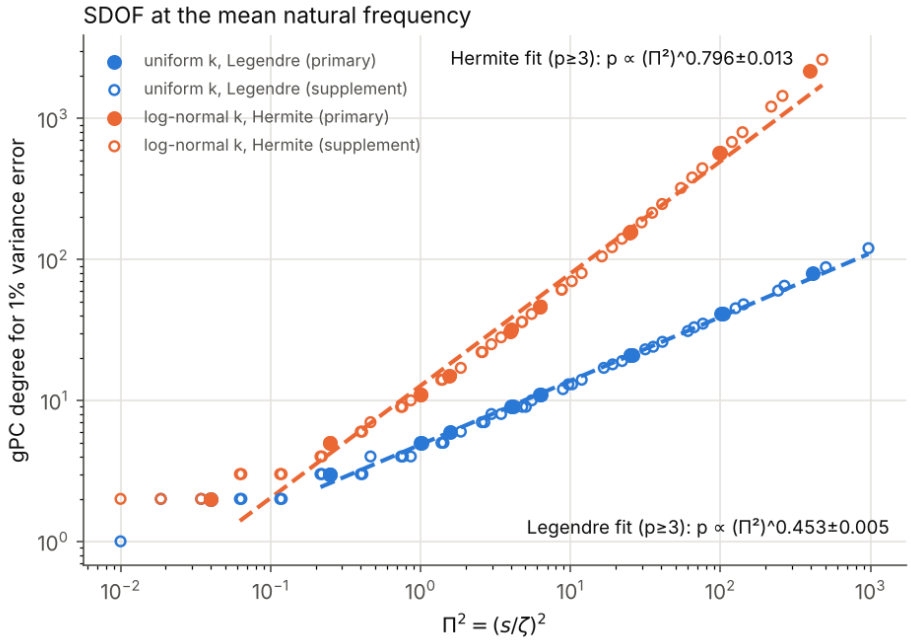
Runner checks: 20-step replay mean rel. L2 = 1.0082898 vs the released train() 1.0082900.

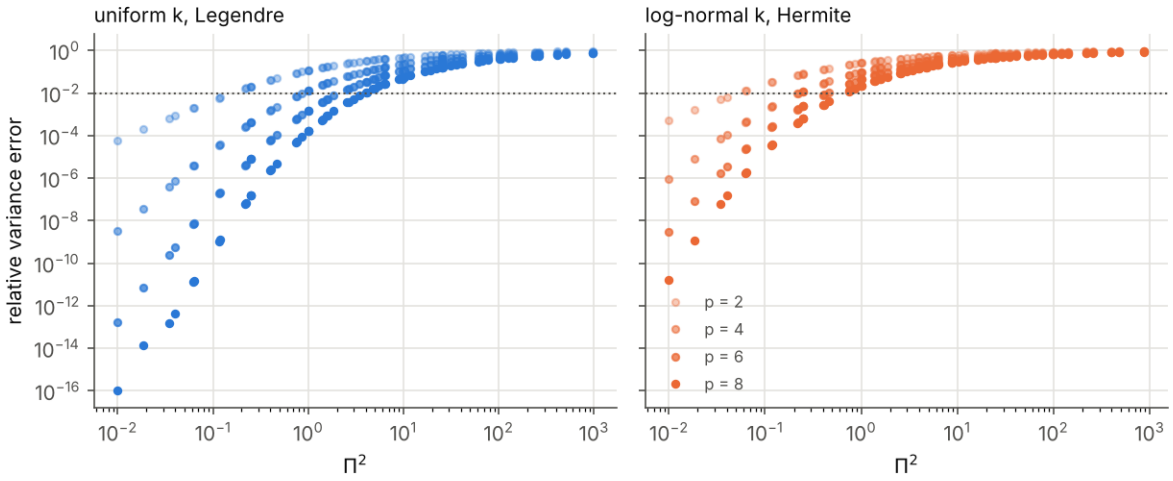
3. (b) H1 on an SDOF oscillator

Receptance of x'' + c x' + k x = $e^{i\omega t}$ with random stiffness (CoV ∈ 2%, 5%, 10%, 20%) and deterministic damper (ζ ∈ 0.5%, 1%, 2%, 5%), plus a 12-point CoV supplement. Required degree = smallest p with relative variance error ≤ 1% (orthonormal projection by quadrature). Π = s/ζ with s the natural-frequency CoV.

Family (input law)	slope of log p vs log Π (p≥3)	R^2	n	local slope by Π bin	p/Π or p/Π² by bin
Legendre (uniform k)	0.905 ± 0.010	0.9939	56	[0.5,2): 0.93 [2,5): 1.02 [5,10): 0.94 [10,40): 0.95	p/Π [0.5,2): 4.69 [2,5): 4.16 [5,10): 4.06 [10,40): 3.98
Hermite (log-normal k)	1.592 ± 0.025	0.9859	59	[0.5,2): 1.41 [2,5): 1.77 [5,10): 1.87 [10,40): 1.93	p/Π² [0.5,2): 10.09 [2,5): 6.92 [5,10): 5.89 [10,40): 5.57

Collapse onto Π: pairs of primary-grid cases with Π within 2% but different ζ and CoV need 9/9, 21/21, 21/21, 41/41, 5/5, 11/11, 21/21, 3/3, 32/31, 157/156, 157/154, 575/567, 11/11, 46/46, 156/154, 5/5 (required degrees).





Reading. The trend is clear, but the exponent depends on the input law. For bounded (uniform/Legendre) inputs, which is how the paper treats θ and W, the required degree grows like Π, not Π². Π² appears only asymptotically for Gaussian (Hermite) inputs. This matches approximation theory (Bernstein ellipse vs strip of analyticity) and the root-spreading result of Jacquelin et al. (2016).

4. (c) Eq. (41): residual-induced damping bias

Derivation (docs/derivations/eq41.md): with a long-window residual loss, an enforced initial condition and a relative unresolved residual floor R, minimising over the decay rate gives $\lambda^{*2} = \lambda_0^2 + (R/2\omega)^2$ with $\lambda_0 = \zeta\omega$. This is Eq. (41) once ζ is read as a decay rate. The steady-state counterpart is ridge shrinkage $X^* = (A^H A + R^2 I)^{-1} A^H F$, a different law.

Pending.

5. (d) Literature check

No published stochastic or separable PINN identifying bearing or support parameters in rotating machinery or a railway wheelset was found (web search only; Crossref and publisher sites are blocked here). Closest: a deterministic PINN for a Jeffcott rotor (Parziale et al., Sensors 24:207, 2024) and gPC-Bayesian magnetic-bearing identification. The near-resonance PCE difficulty itself is established (Jacquelin et al., JSV 2015 and J. Eng. Mech. 2015/2016; Yaghoubi et al., PEM 2017; Kara and Ferguson, JSV 2023), so H1 is framed as a quantified diagnostic. Two full texts (J. Eng. Mech. 2016; PEM 2026) need checking with library access before the scaling law is claimed as new. Full table: docs/lit_review.md.

6. Provenance

Result file	git commit	wall time	timestamp (UTC)
results/p0/s2pinn_cpu_timing.json	unknown	–	2026-10-11T13:21:12Z
results/p0/h1_sdof.json	unknown	0.19 h	2026-10-11T13:38:01Z
results/p0/h1_sdof_summary.json	unknown	–	2026-10-11T13:48:37Z
results/gates/G0.json	3b0f25f2d785…dirty	–	2026-10-11T14:28:13Z